OPTICAL MUSIC RECOGNITION FOR JAZZ LEAD SHEETS: A CRNN-CTC APPROACH FOR MONOPHONIC LINES

Progress Review Milestone

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1 Introduction

This document is the Progress Review of the Bachelor's Thesis *Optical Music Recognition for Jazz Lead Sheets: A CRNN–CTC Approach for Monophonic Lines*, developed under the Computing specialty at FIB–UPC and supervised by Manel Frigola Bourlon (ESAII). It reports the project status at the deadline (21 May 2026) and the plan to defence in June 2026.

The thesis builds an end-to-end OMR system that takes a PDF or image of a jazz lead sheet (Real Book style) and produces a structured transcription of the melody and chord symbols. The core is a CRNN–CTC trained on a synthetic LilyJAZZ corpus derived from PrIMuS, with scan-simulation augmentation and a fine-tuning stage on hand-labelled Real Book pages. The document follows the three sections required by FIB: fulfilment (§2), adjustments (§3), and remaining plan (§4).

2 Fulfilment of the Work Plan and Objectives

2.1 Project Context and Re-Framing

During implementation, the GEP's proof-of-concept scope was refined into an explicit *lead-sheet domain spec* (treble clef only; eight common time signatures; pitch octaves 3–7; monophonic staff; ties; barlines; rests; jazz chord symbols). Vocabulary, dataset filters, grammar fixer, and augmentation tuning all derive from this specification rather than from case-by-case patches.

2.2 Specific Objectives: Status at the Review Date

Table 1 reports the status of each specific objective from the GEP.

2.3 Quantitative Headline Results

The current best checkpoint (`models/latest/best_model.pt`, epoch 37, ResNet-18 backbone) achieves, on the held-out test split of 4 604 samples with greedy CTC decoding:

- Aggregate SER: **1.28 %** on the scanned (augmented) split, **1.19 %** on the clean split.
- Melodic SER (after stripping `measure`, `tied:start`, `tied:stop`): **0.18 %** (scanned), **0.11 %** (clean).
- Perfect-transcription rate: **71 %** (scanned), **73 %** (clean).
- Error budget: ≈74 % of edits are barline (`measure`) tokens and ≈13 % are ties, together ≈87 %. Pitch, octave, and duration error rates are each well below 0.15 %.
- Beam search (width 5) yields only ≤ 1 edit per 1000 SER improvement at roughly 6× decoding cost; greedy is the recommended default and the one used for headline numbers.

These figures meet the implicit GEP target (a CRNN–CTC clearly out-performing the classical baseline). Crucially, residual errors are dominated by structural tokens rather than pitch/duration mistakes.

2.4 Methodology and Rigour

The iterative-incremental methodology from the GEP (two-week cycles, evidence-based replanning, traceability, reproducibility, quantitative control) was followed throughout:

Table 1: Status of the specific objectives at the Progress Review date.

Specific objective (from GEP)	**Status**	**Evidence**
Survey the state of the art (modular, CRNN–CTC, transformers) and justify architectural choices.	Completed	Survey integrated in main thesis; architectural justification consistent with GEP.
Build a baseline pipeline and a CRNN–CTC model for comparison.	Completed (re-scoped)	Classical pipeline implemented for staff detection only; full-symbol baseline de-scoped (see §3).
Construct a training and evaluation dataset using synthetic rendering and augmentation.	Completed	PrIMuS to LilyJAZZ rendering; LMX encoder; albumentations scan-simulation; 87 677 paired samples; ≈270-token vocabulary.
Evaluate model quality with SER and qualitative error analysis.	Completed	Test SER 1.28 % (scanned) / 1.19 % (clean); melodic SER 0.18 % / 0.11 %; 71–73 % perfectly transcribed; per-token error breakdown produced.
Release a reproducible code and configuration workflow.	Completed	Unified `cli.py` (nine subcommands); Poetry environment; `docs/` reference; `Config` serialised inside every checkpoint.
Build a complete inference pipeline (PDF intake, staff detection, melody recognition, grammar fixing, chord OCR) returning structured JSON.	Completed	Five-stage pipeline live; FastAPI exposes a REST endpoint and a browser UI.
Quantify the synthetic-to-real domain gap and reduce it via fine-tuning on Real Book pages.	In progress	Two melody fine-tune cycles done; chord CRNN fine-tuned on hand-labelled strips (last checkpoint 13 May 2026); final per-page real evaluation still pending (see §4).

- **Iterative loop.** Five full data–train–evaluate cycles, each producing a checkpoint under `models/run_*`: cycles 1–3 addressed convergence and vocabulary stability, cycle 4 introduced the augmentation upgrade, cycle 5 integrated rare-token oversampling and header-stripping augmentation.
- **Traceability and reproducibility.** Every checkpoint is paired with a serialised `Config` and a `training_log.csv`; the full workflow is script-driven through `cli.py`.
- **Quantitative control.** SER on the validation split drives early stopping; aggregate and melodic SER are reported separately.
- **Corrective loop.** The most consequential corrections (filter set, vocabulary cleanup, rare-token oversampling, augmentation overlap audit) were enacted at cycle boundaries.

2.5 Analysis of Alternative Solutions

The GEP comparison (traditional modular, CRNN–CTC, transformer end-to-end) holds. Two implementation-time refinements strengthen the original justification: a backbone migration and a redesigned chord pathway. Both are described in detail in §3.

2.6 Integration of Knowledge

The work combines computer vision (preprocessing, morphological staff detection, projection-profile deskewing, augmentation), deep learning (CRNN, CTC, mixed precision, OneCycleLR, beam vs. greedy decoding), music theory and notation (LMX vocabulary, clef/key handling, jazz chord symbology, Real Book engraving), and systems engineering (FastAPI service, hand-labelling UI, Poetry environment, multiprocessing data generation).

A creative cross-domain choice is the use of LilyJAZZ as the synthetic engraving target. Most OMR work trains on classical fonts (Sibelius, default LilyPond), which produces brittle models on Real Book pages. Training directly on LilyJAZZ aligns the synthetic distribution with the inference distribution at no extra cost and is the principal reason the model transfers well after only light fine-tuning.

2.7 Applicable Laws and Regulations

The project does not process personal data and is not subject to GDPR. The PrIMuS dataset is used under its academic-use licence (cited, not redistributed). LilyPond and LilyJAZZ are GNU GPL; the Python stack (PyTorch, OpenCV, albumentations, FastAPI, PyMuPDF) is used under permissive licences tracked in the Poetry lockfile. Real Book pages are kept as a local, non-distributed evaluation set under the educational-use provisions of Spanish copyright law (Ley de Propiedad Intelectual, art. 32); only cropped chord strips paired with text labels are released.

2.8 Commitment, Initiative, and Decision-Making

Work has progressed steadily throughout the academic year, with biweekly supervisor meetings and weekly self-directed sprints. Initiative is reflected in the four documented plan adjustments (§3) and in the introduction of a complementary melodic SER metric, which isolates the pitch/duration error rate that end users actually care about.

Obstacles overcome include: a LilyPond 2.25 API change (`set-global-fonts` removed) that broke the LilyJAZZ stylesheet, patched locally; CTC alignment instability when compressed image width fell below the label length, fixed by `zero_infinity=True` and a tighter multi-staff filter; and augmentation over-darkening from stacking three lighting effects, fixed by an explicit overlap audit.

3 Adjustments to the Plan and their Justification

Execution has followed the GEP plan closely in architecture, methodology, and scope. Four substantive adjustments are documented below; none alter the core objectives and all preserve the 509 h budget.

3.1 Adjustment 1: Chord OCR Pathway Re-Design

The GEP described chord extraction as a generic OCR step (EasyOCR / Tesseract), but Easy-OCR mis-read jazz notation systematically (slash basses as fractions, `maj7` → `moj7`, ∅ unknown) because it is trained on Latin text. The pathway was rebuilt as a dedicated character-level CRNN–CTC reusing the melody training infrastructure: synthetic LilyJAZZ strip renderer, training script, real-strip extractor with pre-labels, browser hand-labelling UI, fine-tuning with weighted sampling, and a grammar-aware post-processor. **Impact:** ≈+35 h, absorbed by the T5.3 buffer and under-spent literature hours.

3.2 Adjustment 2: Backbone Migration from VGG to ResNet-18

The GEP did not pin a CNN backbone. An initial VGG-style stack plateaued at ≈3 % validation SER with mixed-precision instability; migration to ResNet-18 gave roughly 2× faster convergence and stable AMP training. ResNet-18 is now the default; VGG is kept only for old-checkpoint compatibility. **Impact:** absorbed within T3.1/T3.2; no schedule slippage.

3.3 Adjustment 3: Hand-Labelling Tooling

A browser UI (`chord_labeler.html`, `music_labeler.html`) was built into the FastAPI service. It pre-fills the model's prediction, supports keyboard shortcuts, tracks a resumable JSONL ledger, and canonicalises chord-spelling variants (`Cm7` → `C-7`, `m7b5` → ∅). **Impact:** ≈+10 h of UI work, recovered by the labelling speed-up.

3.4 Adjustment 4: Symbol-Level Modular Baseline De-Scoped

T1.3 (full classical symbolic baseline) was scoped down to its morphological staff detector (reused as Stage 2 of inference) plus a qualitative discussion of error propagation. A competitive classical symbol classifier would have cost 40–50 h against an outcome the literature already predicts will lose to the CRNN. **Impact:** −15 h on WP1, redirected to the chord pathway.

3.5 Overall Effect on Plan and Budget

The four adjustments redistribute ≈45 h without changing the total budget (509 h) or the strategic objectives (Table 2).

Table 2: Effort re-allocation across work packages (hours).

Work package	GEP plan	Revised	Delta
WP1 Literature & baseline	78	63	−15
WP2 Dataset construction	73	73	0
WP3 CRNN–CTC development	75	80	+5
WP4 Iterative refinement (incl. chord CRNN)	68	95	+27
WP5 Final evaluation and defence	85	68	−17
WP0 Project management (unchanged)	130	130	0
Total	**509**	**509**	**0**

The economic budget (€16 097.16, dominated by personnel time) and the sustainability matrix (≈57.5 kWh, ≈13.2 kg CO_2eq with the Spanish electricity factor) are unchanged. Actual GPU consumption sits within the projected envelope: five melody training runs and two melody fine-tune cycles, plus three chord training runs and two chord fine-tune cycles, on a single RTX 3060 (12 GB).

4 Work Plan for Completing the Thesis

Approximately five weeks separate the Progress Review deadline (21 May 2026) from the ordinary-period defence in June 2026. Engineering is substantially complete; the remaining effort is predominantly write-up and consolidation.

4.1 Outstanding Engineering Tasks

The following technical items remain open:

E1 – Final melody fine-tune on Real Book staves (8 h)
Run the fine-tune cycle using the full hand-labelled real-staff set; produce the final checkpoint and update `models/latest`.

E2 – Per-page evaluation on real Real Book pages (8 h)
Generate a held-out test of ≈30 real Real Book pages with hand-labelled ground truth; report aggregate SER, melodic SER, perfect-page rate, and per-failure-mode breakdown.

E3 – Final chord-pathway evaluation (4 h)
Re-evaluate the fine-tuned chord CRNN on a held-out set of real strips; compare with the synthetic-only baseline; report character-level and string-level accuracy.

E4 – Final ablations cleanup (4 h)
Wrap up ablation tables: backbone choice, rare-token oversampling, header-stripping augmentation, online jitter.

4.2 Documentation and Write-Up Tasks

The bulk of the remaining effort is the thesis manuscript itself. The current draft has a near-final introduction and state-of-the-art chapter; the remaining chapters are skeleton headings with TODO markers.

W1 – Methodology chapter (18 h)
Data pipeline (rendering, LMX encoding, augmentation), model architecture (ResNet-18 backbone, asymmetric strides, CTC), training loop (AdamW + OneCycleLR + AMP), inference pipeline (five-stage flow, grammar fixer).

W2 – Results chapter (12 h)
SER tables, training curves, error breakdown, domain-gap measurements, fine-tuning effect.

W3 – Management chapter integration (4 h)
Compress GEP material to the main-thesis voice; include the deltas reported in Section 3 of this review.

W4 – Sustainability chapter (3 h)
Carry over and update the sustainability section from the GEP report, with measured (rather than estimated) energy figures where available.

W5 – Conclusions and future work (5 h)
Synthesise findings, discuss limitations, outline polyphony / multi-page / handwritten extensions.

W6 – Front matter and revision pass (5 h)
Abstracts (English/Catalan/Spanish are drafted), revise figures, consistency pass, bibliography sweep.

4.3 Defence Preparation

D1 – Slides (6 h)
20–25 slides; live demo of the FastAPI service on a phone-photographed Real Book page.

D2 – Rehearsal and expected questions (4 h)

4.4 Updated Schedule

Figure 1 shows the weekly plan for the remaining five weeks. Engineering work concludes in week 22; writing dominates weeks 22–25; defence preparation occupies week 25.

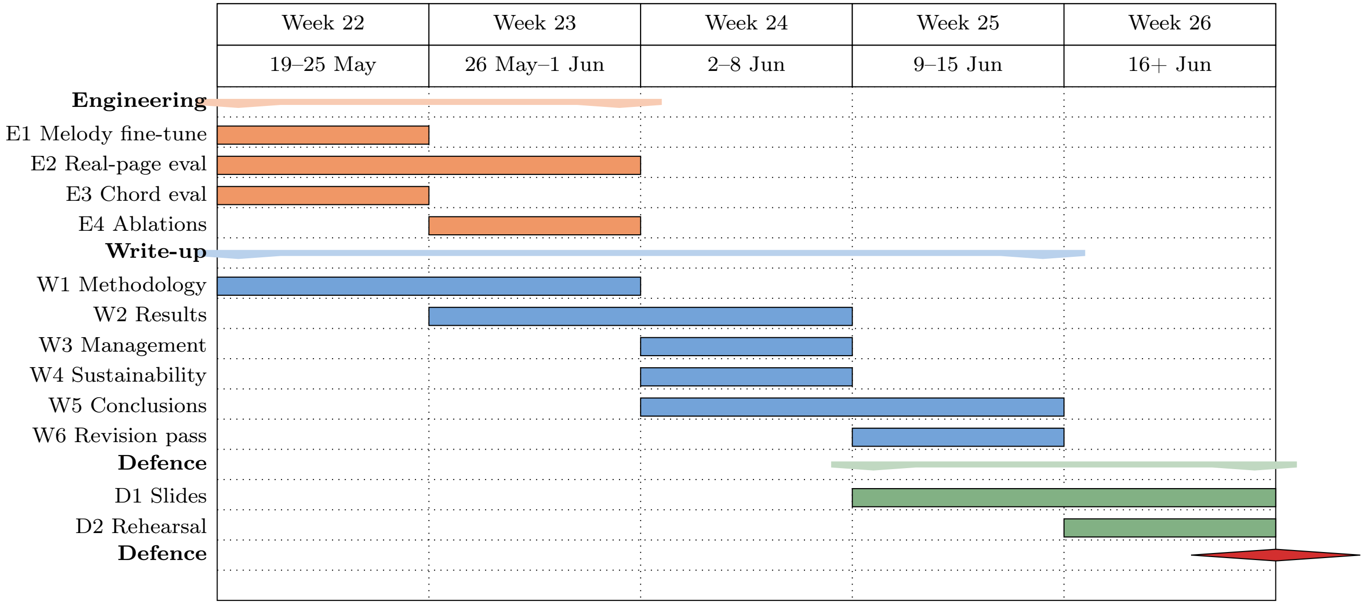

Figure 1: Updated work plan for weeks 22–26 (May–June 2026). Engineering items conclude in week 23; the write-up dominates weeks 22–25; defence preparation is concentrated in weeks 25–26.

4.5 Effort Summary

Table 3 summarises the remaining effort: 81 h across engineering, write-up, and defence. This sits comfortably within a five-week budget of ≈20 h/week.

Table 3: Remaining effort by area.

Area	**Hours**
Engineering (E1–E4)	24
Write-up (W1–W6)	47
Defence preparation (D1–D2)	10
Total remaining	**81**

4.6 Residual Risks

Three residual risks remain:

- **R1 – Real-page SER lags synthetic numbers.** The gap would be reported honestly (melodic SER is the user-relevant metric) and the limitations chapter expanded. Additional fine-tune cycles (≈4 h GPU each) can be triggered if needed.
- **R2 – Write-up overrun.** 47 h in four weeks is feasible but tight. The GEP supplies ≈30 % of the management chapter in reusable form; `docs/` provides the technical scaffold for the methodology chapter.
- **R3 – Defence rehearsal compression.** Slide drafting starts in week 24 in parallel with the conclusions chapter, so rehearsal in week 25 starts from a first-pass deck.

5 Conclusion

The project is on track for defence in June 2026. The core technical objective (a CRNN–CTC for monophonic jazz lead sheets, integrated into an end-to-end pipeline with chord recognition) has been met with measurable margin over the implicit GEP target: 1.28 % aggregate SER and 0.18 % melodic SER on the scanned test split. The four documented plan adjustments reallocate effort without altering scope or budget; they reflect engineering responses to empirical evidence rather than departures from the GEP methodology. Remaining work is principally documentation and defence preparation, both on schedule.